

# AYMANE AIT DADS

Robotics AI Engineer Intern | VLM Integration · Sensor Perception · LLM Orchestration

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## PROFESSIONAL SUMMARY

Ranked 17th out of 3,000+ teams in the NVIDIA Nemotron Reasoning Challenge (score 0.86/1.0, \$106K+ prize pool) by fine-tuning a 30B-parameter model with LoRA SFT on structured multi-category reasoning tasks. Developed a sensor-driven clustering pipeline at Orange Maroc processing 100,000+ fiber-optic network records to classify performance profiles, reducing manual analysis time by 60% through automated feature engineering. Brings hands-on experience with frontier LLMs, prompt engineering, context engineering, and Python-based ML pipelines directly applicable to VLM integration and adaptive task planning for robotic systems. Aims to contribute to AUMOVIO's humanoid robotics R&D in Singapore by building sensor-data processing pipelines and VLM-driven command generation for real-world robotic sorting and manipulation tasks.

## CORE COMPETENCIES

Vision-Language Model Integration | Python Programming | Frontier LLM Landscape | Prompt Engineering | Sensor Perception | LLM Orchestration | Docker | Scene Understanding

## WORK EXPERIENCE

**Orange Maroc** Jul 2024 - Aug 2024

**Data Science Intern** - Casablanca, Morocco

- Built a sensor data processing pipeline ingesting **100,000+ fiber-optic network records** (upload speed, latency, jitter) using Random Forest and K-Means to classify network quality across thousands of nodes.
- Developed a clustering pipeline mapping customer performance profiles to **5 commercial service tiers**, with outputs adopted directly by the network optimization team for infrastructure decisions.
- Reduced manual analysis time by **60%** through automated feature engineering and model evaluation, demonstrating end-to-end pipeline ownership from raw data to actionable output.
- Delivered stakeholder presentations translating model outputs into **maintenance recommendations**, communicating complex technical findings clearly to non-technical decision-makers in English and French.

## PROJECTS

**NVIDIA Nemotron Model Reasoning Challenge** Kaggle Competition · In Progress (June 2026) · Ranked 17th / 3,000+ Teams

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, \$106K+ prize pool) in the ongoing NVIDIA Nemotron Reasoning Challenge by fine-tuning a **30B-parameter Mamba-Transformer** with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context).
- Ran **controlled ablation experiments** across LR schedules, packing strategies, and synthetic augmentation - discovering that reasoning-style consistency outweighs data volume in LLM SFT; built custom cipher parsers and 8-bit operation solvers for verified chain-of-thought data.

PyTorch · Unsloth · PEFT · Hugging Face Transformers · vLLM

**AI-Generated Image Detection - NTIRE 2026 @ CVPR** EURECOM ImSecu + NTIRE 2026 CodaBench · 2025 · Ranked 1st in Class

- Ranked **1st in EURECOM class leaderboard** (0.791 AUC) by fine-tuning CLIP ViT-L/14 (428M parameters) with a custom MLP head, dual learning rates, and 10-view TTA across 250,000 training images from 25+ generator types.
- Also submitted to the **NTIRE 2026 @ CVPR CodaBench** international benchmark; found that 1-epoch training (0.793 AUC) outperformed 4-epoch (0.767 AUC), demonstrating robust out-of-distribution generalization for real-world scene understanding.

PyTorch · OpenCLIP · ViT · FP16 Mixed Precision · Kaggle GPU

## EDUCATION

**Engineering Degree in Data Science (Master-level)** - EURECOM Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Cloud Computing, Image Security, Statistics, Web Applications

**CPGE - Mathematics & Physics** - Ibn Timiya 2021 - 2023

## SKILLS

**AI & LLM Engineering:** Python | Hugging Face Transformers | LoRA / PEFT | LLM Orchestration | Prompt Engineering | vLLM | LangChain

**Computer Vision & ML Frameworks:** PyTorch | OpenCLIP | CLIP ViT | Scikit-learn | TensorFlow | CNN / DenseNet | FP16/BF16 Mixed Precision

**Infrastructure & Data Tools:** Docker | Git | Linux | Pandas | REST APIs | Power BI | PostgreSQL